Chapter 2. Distribution-Free Inference for One Population

Sign test, Wilcoxon signed-rank test, Hodges–Lehmann estimator, efficiency comparisons

Contents

2.1 The sign test **2**
- 2.1.1 The p-value and rejection region of the sign test . . . 3
- 2.1.2 Advantages of the sign test . . . 4
- 2.1.3 Estimating the median . . . 5

2.2 The Wilcoxon signed-rank test **7**
- 2.2.1 Null distribution of the Wilcoxon signed-rank statistic and the p-value . . . 8
- 2.2.2 Large-sample approximation . . . 9
- 2.2.3 Ties . . . 10
- 2.2.4 Estimation and confidence interval for θ via the signed-rank test . . . 12
- 2.2.5 The R function `wilcox.test()` . . . 13

2.3 When should nonparametric methods be used? **14**
- 2.3.1 Comparing inference methods . . . 14
- 2.3.2 Asymptotic relative efficiency of the t-test, sign test and signed-rank test . . . 18

Overview

Using a simple random sample from a population with distribution function $F(x)$, we test hypotheses about the population median $\theta = Q(0.5)$, e.g. the one-sided problem

$$H_0 : \theta = \theta_0, \qquad H_1 : \theta > \theta_0.$$

Example 2.1 (IQ test)

Ten randomly chosen 6th graders of an elementary school received a special one-week training to raise IQ scores, then took an IQ test designed for upper-grade pupils:

$$(98, 121, 110, 89, 109, 108, 102, 92, 131, 114).$$

IQ scores are standardized to mean 100 and standard deviation 10. Is the special training effective?

Example 2.2 (Sodium intake)

The recommended sodium intake for healthy adult men is 1500 mg. The daily intakes of 10 randomly chosen dieting adults were $1200, 2300, 5800, 1100, 1200, 1700, 1400, 1900, 3800, 1300$ (mg). Is the daily intake of dieting adults larger than recommended?

In Example 2.1 the population can be taken normal: with no effect the mean is $\theta = 100$, with an effect $\theta > 100$, so the family is $\{N(\theta, 10) : \theta \in [50, 150]\}$ and the hypotheses are $H_0 : \theta = 100$, $H_1 : \theta > 100$. With known $\sigma = 10$,

$$Z = \frac{\bar{X}_n - \theta_0}{\sigma/\sqrt{n}} = \frac{\bar{X}_n - 100}{10/\sqrt{10}}$$

is standard normal under H_0 (its null distribution). The observed $Z_{\text{obs}} = 7.4$ gives the one-sided p-value $P(Z > 7.4) \approx 6 \times 10^{-14}$, so H_1 (the training is effective) is accepted. In Example 2.2 normality is implausible: the 10 values are right-skewed (long heavy right tail), so even symmetry is hard to assume. When neither normality nor another parametric model is reasonable, distribution-free tests are used.

2.1 The sign test

In Example 2.2 we may use the simplest distribution-free test, the **sign test**. Assume $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ with $F \in \mathcal{F} = \{F\}$, no restriction on the population. Every population has a median $\theta = Q_F(0.5)$, and we test

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0. \tag{2.1}$$

A distribution-free test needs a test statistic whose null distribution does not depend on the actual population. Under H_0 the population can be any distribution with median θ_0; but by definition of the median, half of the population lies below θ_0 and half above. So transform

$$S_i = S_i(X_i, \theta_0) = \begin{cases} +, & X_i > \theta_0, \\ -, & X_i < \theta_0. \end{cases} \tag{2.2}$$

Then $P(S_i = +) = 1/2$ for every population with median θ_0.[1] The S_i are independent, so the number of "+" signs is $B(n, 1/2)$ under H_0. In summary:

[1] More precisely, $P(S_i = +) = P(X_i > \theta_0) = P(X_i \geq \theta_0) = 1/2$ requires $P(X = \theta_0) = 0$ at the median θ_0.

- test statistic of the sign test: $S = \sum_{i=1}^{n} I(S_i = +) = \sum_{i=1}^{n} I(X_i > \theta_0)$;
- null distribution: under H_0, $S \sim B(n, 1/2)$.

Under H_0, $P(X_i > \theta_0) = 1/2$; if in fact $\theta > \theta_0$ (H_1 true), the probability that $S_i = +$ exceeds $1/2$, so S tends to be larger (Figure 2.2 of the book). The distribution-free sign test rejects H_0 when S is large. Since the null distribution is binomial, it is also called the **binomial test**.

2.1.1 The p-value and rejection region of the sign test

The p-value of the sign test is the probability under H_0 of observing $S \sim B(n, 1/2)$ more extreme (in the direction of H_1) than the observed S_{obs}. For the upper-tailed test (2.1), $p = P(S \geq S_{\text{obs}})$.

The upper-tailed test rejects H_0 when S is at least a critical value c_{val}, $S \geq c_{\text{val}}$. To make the type I error at most, and as close as possible to, α, take the smallest c_{val} with

$$P(S \geq c_{\text{val}} \mid H_0) = \sum_{k=c_{\text{val}}}^{n} \binom{n}{k} \left(\frac{1}{2}\right)^n \leq \alpha.$$

In terms of quantiles,

$$c_{\text{val}} = Q_{1-\alpha} + 1, \tag{2.3}$$

where Q_p is the p-quantile of $B(n, 1/2)$.

Proof of (2.3) (supplied). Let F be the distribution function of $B(n, 1/2)$. For an integer c, $P(S \geq c) \leq \alpha \iff 1 - F(c-1) \leq \alpha \iff F(c-1) \geq 1 - \alpha$. By definition $Q_{1-\alpha}$ is the smallest integer x with $F(x) \geq 1 - \alpha$, so the smallest admissible c satisfies $c - 1 = Q_{1-\alpha}$. ■

Example 2.3

In Example 2.1 we test $H_0 : \theta = 100$ against $H_1 : \theta > 100$ (the training is effective) with the sign test. Giving "+" to the observations above $\theta_0 = 100$, $S_{\text{obs}} = 7$ of the $n = 10$ values are positive.

```
> iq = c(98, 121, 110, 89, 109, 108, 102, 92, 131, 114)
> ifelse(iq > 100, "+", "-")
 [1] "-" "+" "+" "-" "+" "+" "+" "-" "+" "+"
```

The null distribution is $S \sim B(10, 1/2)$ and large S_{obs} favors H_1, so the p-value is $P(S \geq S_{\text{obs}}) = \sum_{x=S_{\text{obs}}}^{n} f_B(x; n, \frac{1}{2})$:

```
> (pvalue = 1-pbinom(Sobs-1,n,1/2))
[1] 0.171875
```

(or `sum(dbinom(Sobs:n,n,1/2))`). The p-value is large, so there is not enough evidence that the training is effective.

Example 2.4

With $n = 10$ observations as in Example 2.1, what is the rejection region of the sign test at $\alpha = 0.05$? The distribution of S is discrete, so the type I error cannot be controlled at exactly 0.05. With rejection region $S \geq 9$ the type I error is

```
> S0 = 9
> 1 - pbinom(S0-1, n, 1/2) # P(S >= S0) = 1 - P(S <= S0 -1)
[1] 0.01074219
```

With $S \geq 8$ it exceeds α:

```
> S0 = 8; 1 - pbinom(S0-1, n, 1/2)
```

```
[1] 0.0546875
```

Only with $S \geq 9$ or $S \geq 10$ is the type I error controlled below α, so the rejection region is $S \geq 9$.

Large-sample approximation. The binomial $B(n, p)$ is well approximated by the normal for large n (roughly $np \geq 10$) by the central limit theorem. The null distribution of the sign statistic is $B(n, \frac{1}{2})$, so for large n, under H_0,

$$S \dot{\sim} N\Big(\frac{n}{2}, \frac{n}{2}(1-\tfrac{1}{2})\Big)$$

($X_n \dot{\sim} F$ means the distribution of X_n is approximated by F asymptotically), and the p-value of S_{obs} is approximately

$$p = P(S \geq S_{\text{obs}}) = P\Big(\frac{S - 0.5n}{\sqrt{0.25n}} \geq \frac{S_{\text{obs}} - 0.5n}{\sqrt{0.25n}}\Big) \approx P\Big(Z \geq \frac{S_{\text{obs}} - 0.5n}{\sqrt{0.25n}}\Big),$$

with $Z \sim N(0,1)$; i.e. $p \approx 1 - \Phi\big((S_{\text{obs}} - 0.5n)/\sqrt{0.25n}\big)$. The approximate rejection region at level α is

$$S \geq 0.5n + z_\alpha \sqrt{0.25n}.$$

Example 2.5

The approximate p-value of the sign test for Example 2.1 using the large-sample approximation:

```
> Sz = (Sobs - n/2) / sqrt(0.25*n)
> 1-pnorm(Sz)
[1] 0.1029516
```

With only $n = 10$ observations, however, the large-sample approximation should not be used.

Ties. If X_i is discrete rather than continuous, observations equal to the hypothesized median θ_0 may occur, i.e. a "tie" $X_i = \theta_0$ in (2.2). The simplest treatment deletes tied observations from the test. If m observations equal θ_0, the signs S_i are defined only for the remaining $n' = n - m$, and the null distribution of $S = \sum_{i=1}^{n'} S_i$ is still $B(n', 1/2)$.[2]

Lower-tailed and two-sided tests. Everything above was for the upper-tailed test. For the lower-tailed test $H_0 : \theta = \theta_0$, $H_1 : \theta < \theta_0$, or the two-sided test $H_0 : \theta = \theta_0$, $H_1 : \theta \neq \theta_0$, only the meaning of "extreme" changes. In the lower-tailed test small S_{obs} favors H_1, so $p = P(S \leq S_{\text{obs}}) = \sum_{x=0}^{S_{\text{obs}}} f_B(x; n, \frac{1}{2})$; in the two-sided test, since the null distribution is symmetric, the p-value is twice the smaller one-sided p-value, $p = 2\min\{P(S \geq S_{\text{obs}}), P(S \leq S_{\text{obs}})\}$.

2.1.2 Advantages of the sign test

The sign test is exact under very few assumptions, i.e. over a very wide family of populations. Outliers affect it only through "+" or "−", so it is robust. It needs very simple data: it applies even if the data are only signs.

[2] If X_i is discrete and $P(X_i = \theta_0) = \epsilon > 0$ at the median θ_0, then $P(S_i = +) = P(X_i > \theta_0 \mid X_i \neq \theta_0) = \frac{P(X_i \geq \theta_0) - \epsilon}{1-\epsilon} \geq \frac{.5-\epsilon}{1-\epsilon}$ and $P(S_i = -) = P(X_i < \theta_0 \mid X_i \neq \theta_0) = \frac{P(X_i \leq \theta_0) - \epsilon}{1-\epsilon} \geq \frac{.5-\epsilon}{1-\epsilon}$, so $P(S_i = +) \approx P(S_i = -) \approx 1/2$ and the null distribution of the tie-deleted sign test is approximately $B(n, 1/2)$. If $P(X_i > \theta_0) = P(X_i < \theta_0)$, then $P(S_i = +) = P(S_i = -) = 1/2$ and the null distribution is exactly $B(n, 1/2)$.

What is its advantage over the usual t-test (or Z-test)? To answer we must know in which populations the sign test is more powerful. Let $X_1, \ldots, X_n$ come from a distribution with median θ, and test $H_0 : \theta = \theta_0$ against $H_1 : \theta > \theta_0$ with rejection region $S \geq c_{\text{val}}$. If the true median is $\theta_1 (> \theta_0)$, what is the power? We need the distribution of S under $H_1 : \theta = \theta_1$. Each $I(X_i > \theta_0)$ is Bernoulli with success probability $P(X_i > \theta_0)$, which the value θ_1 alone does not determine; it depends on the distribution of X_i under H_1. E.g. under $\theta = \theta_1$ it differs between $X_i \sim N(\theta_1, 1)$ and $X_i \sim \text{Uniform}(\theta_1 - 1, \theta_1 + 1)$, and so does the power. If the exact alternative distribution is known with $p_1 = P(X_i > \theta_0)$, the power is

$$P(S \geq c_{\text{val}}) = P(S \geq c_{\text{val}} \mid S \sim B(n, p_1)) = 1 - F_B(c_{\text{val}} - 1; n, p_1) \approx 1 - \Phi\Big(\frac{c_{\text{val}} - np_1}{\sqrt{np_1(1-p_1)}}\Big),$$

the last by the large-sample approximation.

Example 2.6

For the data of Example 2.1 assume $X_i \sim N(\theta, 10^2)$, and compare the power of two tests of $H_0 : \theta = 100$ versus $H_1 : \theta > 100$:

(a) Z-test with rejection region $\bar{X}_n > 100 + z_{0.05}10/\sqrt{10}$ (type I error 5%);

(b) sign test with rejection region $S \geq 8$ (type I error 5.5%).

If in fact $\theta = 105$, what is the probability that each test accepts H_1 (the power)? For the Z-test, under $\theta = 105$, $Z_n = (\bar{X}_n - 105)/\sqrt{10} \sim N(0, 1)$, so

$$P(\bar{X}_n > 100 + z_{0.05}10/\sqrt{10}) = P\Big(Z_n > \frac{-5 + z_{0.05}\sqrt{10}}{\sqrt{10}}\Big) \approx 0.47, \tag{2.4}$$

a power of 47%. For the sign test, under $\theta = 105$, $X_i \sim N(105, 10^2)$ and each sign is + with probability $p_1 = P(X_i > 100) = 1 - \Phi(-1/2) \approx 0.69$. So $S \sim B(n, 0.69)$ and the power of the sign test is $P(S \geq 8) \approx 0.36$. When the population really is normal, the Z-test is much more powerful.

The sign test can be more powerful than the Z-test when the population is not normal: for the heavy-tailed Laplace distribution, the sign test beats the Z-test.

Example 2.7 (Example 2.6 continued)

Suppose the alternative population is Laplace$(105, \sqrt{50})$ (the scale $\sqrt{50}$ makes the standard deviation 10). The Z-statistic $\bar{X}_n$ is no longer exactly normal, but the Laplace is symmetric, so by the CLT $\bar{X}_n$ is well approximated by a normal, and the power of the Z-test is approximately 0.47 (from (2.4)). For the sign test, $p_1 = P(X_i > 100 \mid X_i \sim \text{Laplace}(105, \sqrt{50})) \approx 0.75$, giving

$$P(S \geq 8) \approx 0.54,$$

higher than the Z-test.

In summary, the sign test is more powerful, hence more efficient, for heavy-tailed populations.

2.1.3 Estimating the median

Using the sample median $\hat{\theta} = \text{median}\{X_1, \ldots, X_n\}$ and order statistics, we obtain an estimator and a distribution-free confidence interval for the population median θ. Because the sign test is distribution-free, the estimator associated with it is the sample median (the *Hodges–Lehmann estimator* of the sign test, see below), and the order-statistic interval keeps its confidence level whatever the population.

Confidence interval. Let $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$ be the order statistics. They relate simply to the sign statistic $S_\theta = \sum_{i=1}^n I(X_i > \theta)$: the number of observations larger than θ is S_θ. For a confidence level $1-\alpha$ we look for order statistics $(X_{(l)}, X_{(u)})$ such that the median θ lies in

$$X_{(l)} \le \theta < X_{(u)} \tag{2.5}$$

with probability at least $1-\alpha$. $X_{(l)} \le \theta$ means at most $n-l$ observations exceed θ, and $\theta < X_{(u)}$ means at least $n+1-u$ observations exceed θ.[3] Hence (2.5) is equivalent to

$$n-u+1 \le \sum_{i=1}^{n} I(X_i > \theta) \le n-l,$$

and since $S_\theta = \sum I(X_i > \theta) \sim B(n, 1/2)$ we seek (l, u) with

$$P(n-u+1 \le S_\theta \le n-l) = \sum_{x=n-u+1}^{n-l} \binom{n}{x} (0.5)^n = \sum_{x=l}^{u-1} \binom{n}{x} (0.5)^n \ge 1-\alpha \tag{2.6}$$

of smallest length $(u-1)-l$. For intervals with both endpoints, it is customary to make the probabilities on both sides outside the interval equal: the probability $P(\theta \ge X_{(u)})$ that θ exceeds the upper limit and $P(\theta < X_{(l)})$ that it is below the lower limit are forced to be equal,

$$P(\theta \ge X_{(u)}) = P(\theta < X_{(l)}) \le \alpha/2. \tag{2.7}$$

Then $P(X_{(l)} \le \theta < X_{(u)}) = 1 - P(\theta \ge X_{(u)}) - P(\theta < X_{(l)}) \ge 1-\alpha$. The events in (2.7) are $\{S_\theta \le n-u\}$ and $\{S_\theta \ge n-l+1\}$, and by the symmetry of $B(n, \frac{1}{2})$, $P(S_\theta \le n-u) = P(S_\theta \ge u)$ and $P(S_\theta \ge n-l+1) = P(S_\theta \le l-1)$. So (2.7) reads

$$P(S_\theta \ge u) = P(S_\theta \le l-1) \le \alpha/2,$$

which forces $u = n-l+1$. Therefore take the largest l with $P(S_\theta \le l-1) = P(S_\theta < l) \le \alpha/2$,

$$l := Q_B(\alpha/2).$$

With this,[4] the $1-\alpha$ confidence interval is $X_{(Q_B(\alpha/2))} \le \theta < X_{(n-Q_B(\alpha/2)+1)}$.[5]

Example 2.8

The 95% confidence interval for the median sodium intake of Example 2.2:

```
> x = sort(c(1200, 2300, 5800, 1100, 1200, 1700, 1400, 1900,
    3800, 1300))
> n = length(x)
> alpha = 0.05
> l = qbinom(alpha/2, size = n, prob = 1/2)
> u = n + 1 - l
> (ci = x[c(l,u)])
[1] 1200 3800
```

The coverage probability of $(1200, 3800)$ exceeds 95%:

```
> (coverage = sum(dbinom(l:(u-1), n, prob = 1/2)) )
[1] 0.9785156
```

[3] The inequalities in (2.5) were chosen so that the interval is equivalent to a condition on S_θ. If the population is continuous, $P(X_{(l)} = \theta) = 0$, so open and closed intervals have the same confidence level.

[4] More precisely, if $P(S_\theta \le Q_B(\alpha/2)) > \alpha/2$ then $l = Q_B(\alpha/2)$; if $P(S_\theta \le Q_B(\alpha/2)) = \alpha/2$ then $l = Q_B(\alpha/2)+1$. For the usual levels ($\alpha = 0.05$ or 0.01) the second case essentially never occurs, so $l = Q_B(\alpha/2)$ is fine.

[5] Intervals like (2.5) can also be obtained from the duality between tests and confidence intervals; see Song et al. (2015), Section 2.4.

For large n an approximate interval uses the large-sample approximation $S_\theta \dot{\sim} N(0.5n, 0.25n)$ in (2.6): find integers l^*, u^* between 1 and n with

$$\frac{l - 0.5n}{\sqrt{0.25n}} \approx -z_{\alpha/2}, \qquad \frac{u - 1 - 0.5n}{\sqrt{0.25n}} \approx z_{\alpha/2}.$$

The normal-approximation interval $(X_{(l^*)}, X_{(u^*)})$ is also distribution-free and asymptotically valid.

Inference for quantiles. The sign test for the median extends naturally to quantiles. E.g. test the 10% quantile $\theta_{0.1} = Q(0.1)$. Under $H_0 : \theta_{0.1} = \theta_0$, the sign of $X_i - \theta_0$ is $-$ with probability 1/10 and $+$ with probability 9/10. So the null distribution of $S_{\theta_0} = \sum_{i=1}^n I(X_i > \theta_0)$ is $B(n, 0.9)$, and the lower-tailed test $(H_1 : \theta_{0.1} < \theta_0)$ has p-value $p = P(S_{\theta_0} \le S_{\text{obs}}) = F_B(S_{\text{obs}}; n, 0.9)$.

A distribution-free confidence interval for the p-quantile $Q(p)$ also uses order statistics. Let $N_\theta = \#\{i : X_i \le \theta\} \sim B(n, p)$ at $\theta = Q(p)$ (continuous population). As before, $X_{(l)} \le Q(p) < X_{(u)} \iff l \le N_\theta \le u - 1$, so we seek (l, u) with

$$P(l \le N_\theta \le u - 1) = \sum_{x=l}^{u-1} \binom{n}{x} p^x (1-p)^{n-x} \ge 1 - \alpha.$$

(For $p = 1/2$ this is (2.6).)

2.2 The Wilcoxon signed-rank test

If the population f can be assumed symmetric about the median θ, there is a more powerful test than the sign test. The population then belongs to

$$f \in \mathcal{F}_2 = \{f : f(\theta - x) = f(\theta + x),\ x \in (-\infty, \infty) \text{ for some } \theta \in (-\infty, \infty)\},$$

and if the mean exists it equals the median. Under this assumption consider the two-sided test: with $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} f \in \mathcal{F}_2$, for the median θ,

$$H_0 : \theta = \theta_0, \qquad H_1 : \theta \ne \theta_0. \tag{2.8}$$

Wilcoxon (1945) extracted more information from the data than the signs, expecting a more efficient (more powerful) test. The **signed-rank test statistic** uses both signs and ranks relative to θ_0. The sign is $I(X_i - \theta_0)$ (i.e. 1 if $X_i > \theta_0$, 0 if $X_i < \theta_0$; observations with $X_i = \theta_0$ are deleted), and the rank R_i is the position of the absolute deviation $D_i = |X_i - \theta_0|$ in increasing order: if the ith absolute deviation is the i'th smallest $(D_i = D_{(i')})$, then $R_i = i' \in \{1, \ldots, n\}$.

Example 2.9

For observations $11, 20, 16, 5, 3, 17$ and $\theta_0 = 15$, Wilcoxon's signs and ranks are:

X_i	11	20	16	5	3	17
deviation $X_i - \theta_0$	-4	$+5$	$+1$	-10	-12	$+2$
sign S_i	$-$	$+$	$+$	$-$	$-$	$+$
sign $I(X_i > \theta_0)$	0	1	1	0	0	1
absolute deviation D_i	4	5	1	10	12	2
rank R_i	3	**4**	**1**	5	6	**2**

Wilcoxon's signed-rank statistic is the sum of the ranks with positive sign,

$$SR_+ = \sum_{i=1}^n I(X_i > \theta_0) \cdot R_i.$$

If the null hypothesis "symmetric about θ_0" is true, then given that the absolute deviation $|X_i - \theta_0|$ equals d, the deviation is positive or negative with exactly equal probability:

$$P(X_i > \theta_0 \mid |X_i - \theta_0| = d) = \frac{1}{2} = P(X_i < \theta_0 \mid |X_i - \theta_0| = d). \tag{2.9}$$

So under H_0, $I(X_i > \theta_0)R_i$ and $I(X_i < \theta_0)R_i$ have the same expectation, and SR_+ should be neither too large nor too small. If $\theta > \theta_0$, deviations of larger rank tend to be positive, so SR_+ is large; if $\theta < \theta_0$, SR_+ is small. For the two-sided alternative we reject H_0 when SR_+ is too large or too small.

In Example 2.9 the ranks with positive sign are $1, 2, 4$, so $SR_+ = 1 + 2 + 4 = 7$. This is smaller than the negative ranks $3, 5, 6$, but is it small enough to reject H_0? The answer comes from the null distribution of the statistic, which also gives the p-value and the rejection region.

Example 2.10

In R the signed-rank statistic for the data of Example 2.9:

```
> x = c(11,20,16,5,3,17); theta0 = 15
> D = x - theta0 # deviations
> sign(D) # signs
[1] -1  1  1 -1 -1  1
> rank(abs(D)) # ranks of absolute deviations
[1] 3 4 1 5 6 2
> (SR = sum( ( sign(D)>0 ) * rank(abs(D)) ) )
[1] 7
```

2.2.1 Null distribution of the Wilcoxon signed-rank statistic and the p-value

By (2.9), under H_0 each rank R_i has a positive or negative sign with probability $1/2$. The key observation is that the ranks $1, 2, \ldots, 6$ are regarded as fixed (whatever continuous population the observations come from, the ranks are always $1, 2, \ldots, 6$). Under H_0 each rank independently gets a positive sign with probability $1/2$, i.e. enters SR_+ with probability $1/2$. So the signed ranks S_iR_i take the

$$(\pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6) \tag{2.10}$$

$2^6 = 64$ possible values, each sign arising with probability $1/2$, so each configuration in (2.10) has probability $(1/2)^6 = 1/64$. All cases under H_0:

(signed ranks)	ranks in SR_+	SR_+	probability
$(-1, -2, -3, -4, -5, -6)$	none	0	1/64
$(+1, -2, -3, -4, -5, -6)$	1	1	1/64
$(-1, +2, -3, -4, -5, -6)$	2	2	1/64
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$(+1, +2, -3, +4, -5, -6)$	1,2,4	7	1/64
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$(+1, +2, +3, +4, +5, +6)$	1,2,3,4,5,6	21	1/64

The last two columns give the null distribution of SR_+; tabulating gives its probability mass function (Figure 2.3 of the book: $n = 6$, no ties, symmetric about 10.5).

For Example 2.9, the probability of a value at most the observed $SR_+ = 7$ is, from the null distribution, $P(SR_+ \leq 7) = 18/64 \approx 0.28$. This is the p-value of the lower-tailed test; since the

null distribution is symmetric, the p-value of the two-sided test (2.8) is 0.56, and $H_0 : \theta = \theta_0$ is not rejected.

To derive the null distribution of SR_+ we fixed the ranks R_i and resampled only the signs $S_i = +, -$; this is generally called the **re-randomization method**. The null distribution of SR_+ depends only on the sample size n and is not a well-known parametric distribution (whereas the sign statistic is binomial). Since it is used often, R's `dsignrank()` gives its probability mass function; it is called **Wilcoxon's signed-rank distribution**.

To get the rough shape of the null distribution, compute its mean and variance. Under H_0 the $I(X_i > \theta_0)$ are independent, equal to 1 or 0 with probability 1/2 each; with $V_i \equiv I(X_i > \theta_0)$, $V_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(1/2)$ and

$$SR_+ = \sum_{i=1}^{n} V_i R_i \overset{d}{=} \sum_{i=1}^{n} i V_i. \tag{2.11}$$

Justification of (2.11) (supplied). Under H_0, by (2.9) the sign vector $(V_1, \ldots, V_n)$ is independent of the absolute deviations $(D_1, \ldots, D_n)$, hence of the ranks, and consists of i.i.d. Bernoulli(1/2) variables. Given the ranks, $(R_1, \ldots, R_n)$ is a permutation π of $(1, \ldots, n)$, and $\sum_i V_i R_i = \sum_k k V_{\pi^{-1}(k)}$, where $(V_{\pi^{-1}(k)})_k$ is again i.i.d. Bernoulli(1/2). So the conditional distribution given the ranks is that of $\sum_k k V_k$, and so is the unconditional one. ■

Hence

$$E(SR_+) = \sum_{i=1}^{n} i E(V_i) = \frac{1}{2} \sum_{i=1}^{n} i = \frac{n(n+1)}{4},$$

$$\text{Var}(SR_+) = \sum_{i=1}^{n} i^2 \,\text{Var}(V_i) = \frac{1}{4} \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{24},$$

and the distribution is symmetric about the mean $n(n+1)/4$ (replacing V_i by $1 - V_i$, which has the same law, maps SR_+ to $\frac{n(n+1)}{2} - SR_+$). With observed value SR_{obs} from a sample of size n, the p-values are:

- upper alternative $H_1 : \theta > \theta_0$: $p_r = P(SR_+ \geq SR_{\text{obs}})$;
- lower alternative $H_1 : \theta < \theta_0$: $p_l = P(SR_+ \leq SR_{\text{obs}}) = F_W(SR_{\text{obs}}; n)$;
- two-sided $H_1 : \theta \neq \theta_0$: by symmetry of the signed-rank distribution, $p = 2 \min\{p_r, p_l\}$.

2.2.2 Large-sample approximation

For large n, computing SR_+ for all 2^n signed-rank configurations is infeasible. Then the mean and variance give the standardized statistic, approximately standard normal for large n:

$$Z := \frac{SR_+ - E(SR_+)}{\sqrt{\text{Var}(SR_+)}} = \frac{SR_+ - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}} \overset{\cdot}{\sim} N(0,1). \tag{2.12}$$

Proof of (2.12). The normal approximation is the central limit theorem, as for the sign statistic S. By (2.11), $SR_+ = \sum_i i V_i = \sum_i U_i$, where under H_0 the $U_i = i V_i$ are independent with

$$U_i = \begin{cases} 0 & \text{with probability } 1/2, \\ i & \text{with probability } 1/2. \end{cases}$$

The U_i are not identically distributed, so we use Lyapunov's central limit theorem.[6] (Supplied verification.) Take $\delta = 1$: $|U_i - i/2| = i/2$ always, so $\sum_i E|U_i - \mu_i|^3 = \sum_i i^3/8 = O(n^4)$, while

[6] Lyapunov's CLT: let $X_1, \ldots, X_n$ be independent with means μ_i and variances σ_i^2, and $s_n^2 = \sum_{i=1}^n \sigma_i^2$. If for some $\delta > 0$ Lyapunov's condition $\lim_{n\to\infty} \frac{1}{s_n^{2+\delta}} \sum_{i=1}^n E\big[|X_i - \mu_i|^{2+\delta}\big] = 0$ holds, then $s_n^{-1} \sum_i (X_i - \mu_i) \xrightarrow{d} N(0,1)$.

$s_n^2 = n(n+1)(2n+1)/24$ gives $s_n^3 \asymp n^{9/2}$. The ratio is $O(n^{-1/2}) \to 0$, so Lyapunov's condition holds and $Z \xrightarrow{d} N(0,1)$. ■

The approximate p-value of the upper-tailed test is

$$P(SR_+ \geq SR_{\text{obs}}) = P\Big(Z \geq \frac{SR_+ - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}}\Big) \approx 1 - \Phi\Big(\frac{SR_{\text{obs}} - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}}\Big). \quad (2.13)$$

Example 2.11

To see whether counseling changes depression scores of 12 patients, scores before counseling were (in patient-ID order) $(1.83, 0.5, 1.62, 2.48, 1.68, 1.88, 1.55, 3.06, 1.30, 0.39, 0.94, 1.34)$ and after counseling $(0.88, 0.65, 0.59, 2.05, 1.06, 1.29, 1.06, 3.14, 1.29, 0.43, 0.57, 0.92)$. To see whether counseling lowers the score, run the signed-rank test with the large-sample approximation. For paired tests, record the differences of the pairs as a new vector `x`; large values mean the depression score decreased. This is an upper-tailed test, so use (2.13):

```
> x.before = c(1.83, 0.5, 1.62, 2.48, 1.68, 1.88, 1.55, 3.06,
    1.30, 0.39, 0.94, 1.34)
> x.after = c(0.88, 0.65, 0.59, 2.05, 1.06, 1.29, 1.06, 3.14,
    1.29, 0.43, 0.57, 0.92)
> x = x.before - x.after
> D = x - 0
> (SR = sum( ( sign(D)>0 ) * rank(abs(D)) ) )
[1] 69
> n = length(x)
> Z_SR = (SR - n*(n+1)/4)/ sqrt(n*(n+1)*(2*n+1)/24)
> (p = 1-pnorm(Z_SR))
[1] 0.009301465
```

The p-value is small: there is substantial evidence that counseling lowers depression scores.

2.2.3 Ties

Example 2.12

For 10 healthy volunteers we want to know whether blood pressure rises after exercise compared with before. After an hour of high-intensity exercise the differences in blood pressure were $-3, -2, -2, 0, 1, 2, 4, 8, 11, 11$. Assuming the differences are symmetric about the median θ, we test whether the median difference exceeds 0.

These data have two kinds of ties:

- observations equal to the hypothesized median θ_0 (here 0);
- observations with equal absolute deviations: here $-2, -2, 2$ all have $D_i = |X_i - \theta_0| = 2$, and $11, 11$ must get the same rank.

The first kind is handled as in the sign test: delete the observation and take the sample size as 9. For the second kind one usually uses **mid-ranks**. Exactly one absolute deviation is smaller than 2, so the three tied values would occupy ranks $2, 3, 4$; since they cannot be ordered, each gets the average $3 = \frac{2+3+4}{3}$. The signed-rank statistic with ties is then computed from:

X_i	-3	-2	-2	0	1	2	4	8	11	11
deviation $X_i - \theta_0$	-3	-2	-2	0	1	2	4	8	11	11
sign S_i	$-$	$-$	$-$		$+$	$+$	$+$	$+$	$+$	$+$
sign $I(X_i > \theta_0)$	0	0	0		1	1	1	1	1	1
absolute deviation D_i	3	2	2		1	2	4	8	11	11
rank R_i	5	3	3		$\mathbf{1}$	$\mathbf{3}$	$\mathbf{6}$	$\mathbf{7}$	$\mathbf{8.5}$	$\mathbf{8.5}$

The signed-rank statistic is the sum of positive ranks, $SR_+ = 1 + 3 + 6 + 7 + 8.5 + 8.5 = 34$. (The book prints 39, an arithmetic slip; the R output below confirms the ranks.) Its null distribution differs from the tie-free case: the R_i are not the integers $1, 2, \ldots, n$ but $1, 3, 3, 3, 5, 6, 7, 8.5, 8.5$, so the last equality of (2.11) fails. The re-randomization principle still applies: under H_0, fix the ranks R_i and let the signs $V_i = I(S_i = +)$ be independent with probability 1 or 0 each, $V_i \overset{\text{i.i.d.}}{\sim}$ Bernoulli(1/2); with all 2^n cases, re-randomization gives the null distribution.

Example 2.13

R code deriving the null distribution by re-randomization with ties, for testing whether the median of the data below is 0:

```
> toy.data = c(-3,-2,-2,0,1,2,4,8,11,11); theta0 = 0
> dev = toy.data - theta0
> reduced.dev = dev[sign(dev) != 0 ]
> (ranks = rank(abs(reduced.dev)))
[1] 5.0 3.0 3.0 1.0 3.0 6.0 7.0 8.5 8.5
```

All sign configurations are generated by `permutations()` of the `gtools` package: the rows of `allsignflips` are all length-n vectors with entries in `v = c(1,-1)` (with repetition, `repeats.allowed = TRUE`); there are 2^n rows.

```
> n = length(ranks)
> allsignflips = gtools::permutations(2, n, v = c(1,-1),
+     repeats.allowed = TRUE)
> nperm = nrow(allsignflips) # equals 2^n
```

Compute the rank sum for each of the 2^n sign flips:

```
> SRvec = vector(length = nperm) # rank sum in each case
> for (i in 1:nperm){
+   SRvec[i] = sum(ranks[allsignflips[i,] > 0])
+ }
```

The number of sign flips giving a rank sum is its frequency; dividing by 2^n gives the probability mass function of the null distribution.

```
> N = n*(n+1)/2
> dsignrank.tie = vector(length = N+1)
> for (sr in 0:N){
+   dsignrank.tie[sr+1] = sum(SRvec == sr) / nperm
+ }
> data.frame(SR = 0:N, SRdensity = dsignrank.tie) %>% head(3)
  SR   SRdensity
1  0 0.001953125
2  1 0.001953125
3  2 0.000000000
```

Because of the ties no rank equals 2, so $SR_+ = 2$ has probability 0. By contrast `dsignrank()` cannot produce the exact null distribution with ties:

```
> dsignrank(0:2,n)
```

```
[1] 0.001953125 0.001953125 0.001953125
```

2.2.4 Estimation and confidence interval for θ via the signed-rank test

For observations $X_1, \ldots, X_n$, the averages of all pairs

$$W_{ij} := \frac{X_i + X_j}{2}, \qquad 1 \le i \le j \le n,$$

are the **Walsh averages**. For $i = j$, $W_{ii} = X_i$; there are $N = n(n+1)/2$ Walsh averages. The number of Walsh averages larger than a given value θ equals the signed-rank statistic for hypothesized median θ:

$$SR_+(\theta) = \#\{W_{ij} > \theta\}. \tag{2.14}$$

To see this roughly, suppose the observations are ordered and distinct ($X_1 < X_2 < \cdots < X_n$, i.e. $X_i = X_{(i)}$). For $i < j$, $W_{ij} < W_{i,j+1}$ and $W_{ij} < W_{i+1,j}$; and $W_{ij} > \theta$ means X_i is closer to θ than X_j. The largest Walsh average is $W_{nn} = X_n$, the second is $W_{n-1,n}$, and the third is $W_{n-1,n-1} = X_{n-1}$ or $W_{n-2,n}$. Arrange the Walsh averages from W_{11} to W_{nn}. If $\theta > W_{nn}$, clearly $SR_+ = 0 = \#\{W_{ij} > \theta\}$. Next, if $\theta \in (W_{n-1,n}, W_{nn})$ then $\#\{W_{ij} > \theta\} = 1$, and the only observation above θ is X_n with $|X_n - \theta| < |X_{n-1} - \theta|$, so $SR_+ = 1$. By the same logic, each time θ passes a Walsh average W_{ij}, SR_+ increases by 1.

Proof of (2.14) (supplied). Assume no ties among the $|X_k - \theta|$ and no W_{ij} equal to θ, and let $d_k = X_k - \theta$. Then $W_{ij} > \theta \iff d_i + d_j > 0$. Assign each pair $\{i, j\}$ $(i \le j)$ to the index $k \in \{i, j\}$ with the larger $|d_k|$. The sign of $d_i + d_j$ is the sign of d_k, since the other term has smaller absolute value (or is d_k itself if $i = j$). The pairs assigned to k are $\{k, m\}$ with $|d_m| \le |d_k|$, i.e. exactly R_k pairs, where R_k is the rank of $|d_k|$. Hence

$$\#\{W_{ij} > \theta\} = \sum_{k:d_k>0} R_k = SR_+(\theta).$$

■

By (2.14), the signed-rank statistic is the sign statistic of the Walsh averages (# of W_{ij} larger than θ). Just as the sign test (number of observations above θ) gave intervals $X_{(l)} \le \theta < X_{(u)}$, the signed-rank test gives intervals from the order statistics $W_{(1)} < \cdots < W_{(n(n+1)/2)}$ of the Walsh averages.[7]

For a confidence level $1 - \alpha$ we take order statistics $W_{(l)}, W_{(u)}$ such that the median θ lies in

$$W_{(l)} \le \theta < W_{(u)} \tag{2.15}$$

with probability at least $1 - \alpha$. As for the sign test, choose l, u so that the two outside probabilities are equal; each must be at most $\alpha/2$ for coverage $1 - \alpha$:

$$P(\theta \ge W_{(u)}) = P(\theta < W_{(l)}) \le \frac{\alpha}{2}. \tag{2.16}$$

The distribution of the Walsh averages depends on the population, so instead use the following relation, which allows the Wilcoxon signed-rank distribution of SR_+ with the unknown population: with $N = n(n+1)/2$ and $a \in \{1, \ldots, N\}$,

$$\begin{aligned} SR_+ > N - a &\iff \theta < W_{(a)}, \\ SR_+ \le N - a &\iff \theta \ge W_{(a)}. \end{aligned} \tag{2.17}$$

[7] This interval can be justified by inverting the test. Whether $H_0 : \theta = \theta_0$ is rejected depends on $SR_+(\theta_0)$, i.e. on the relative position of θ_0 among the Walsh averages; the values θ_0 not rejected at level α form the $1 - \alpha$ interval. Both methods give the same interval.

(By (2.14), $SR_+(\theta) > N - a$ means more than $N - a$ Walsh averages exceed θ, i.e. $\theta < W_{(a)}$.) The distribution of SR_+ is symmetric about $n(n+1)/4$, and by (2.17), (2.16) becomes

$$P(SR_+ \geq u) = P(SR_+ \leq l - 1) \leq \frac{\alpha}{2}.$$

Here $u = n(n+1)/2 - l + 1$. So with the largest l satisfying $P(SR_+ \leq l - 1) = P(SR_+ < l) \leq \alpha/2$,

$$l := Q_{SR}(\alpha/2),$$

the $1 - \alpha$ interval is $W_{(Q_{SR}(\alpha/2))} \leq \theta < W_{(n(n+1)/2 - Q_{SR}(\alpha/2)+1)}$. As the confidence level shrinks, what remains is the median of the Walsh averages,

$$\hat{\theta} = \text{median}\{W_{(ij)}\},$$

an estimator of the population median θ: the **Hodges–Lehmann estimator**.[8]

Example 2.14

For Example 2.11, as an answer to how much counseling lowers depression scores, compute the nonparametric estimate of the median θ of the score differences and a 95% confidence interval, assuming the population is symmetric about its median θ. The Walsh averages come from `Rfit::walsh()`.

```
> w = sort(Rfit::walsh(x)) # sorted walsh averages
> median(w)
[1] 0.4225
> alpha = 0.05
> u = n*(n+1)/2 + 1 - qsignrank(alpha/2,n)
> l = qsignrank(alpha/2,n)
> c(l,u)
[1] 14 65
> (ci = c(w[l], w[u]))
[1] 0.145 0.685
```

The median estimate is 0.4225 and the 95% interval is $(0.145, 0.685)$.

2.2.5 The R function `wilcox.test()`

The signed-rank test is implemented in R's `wilcox.test()`. For the counseling data of Example 2.11, with the differences in vector `x`: `alternative = "greater"` gives the upper-tailed test, `exact = F` the large-sample approximation, and `correct = F` turns off the continuity correction used in normal approximations of discrete distributions.

```
> wilcox.test(x, alternative = "greater", exact = F, correct =F)

        Wilcoxon signed rank test

data:  x
V = 69, p-value = 0.009301
alternative hypothesis: true location is greater than 0
```

This matches Example 2.11. For the exact test using the exact null distribution of the signed-rank statistic, set `exact = TRUE` (exact only without ties); no continuity correction is needed.

[8] Hodges and Lehmann (1963) proposed obtaining estimators of a location parameter from tests; these are called Hodges–Lehmann estimators. The Hodges–Lehmann estimator from the signed-rank test is the median of the Walsh averages; from the sign test it is the sample median. It was also proposed independently by Sen (1963), and is sometimes called the Hodges–Lehmann–Sen estimator.

```
> wilcox.test(x, alternative = "greater", exact = T)

        Wilcoxon signed rank exact test

data:  x
V = 69, p-value = 0.008057
alternative hypothesis: true location is greater than 0
```

The two-sided test and the accompanying Walsh-average interval:

```
> wilcox.test(x, conf.int = T)

        Wilcoxon signed rank exact test

data:  x
V = 69, p-value = 0.01611
alternative hypothesis: true location is not equal to 0
95 percent confidence interval:
 0.145 0.685
sample estimates:
(pseudo)median
        0.4225
```

This matches Example 2.14.

2.3 When should nonparametric methods be used?

If the population is approximately normal, or the sample mean is approximately normal by the CLT, the Z- or t-test gives inference on the mean. In comparison, the sign test is exact and distribution-free for any population, and the signed-rank test is exact when the population is symmetric. Should we then always prefer the signed-rank test over the sign test for symmetric populations? Relatedly, may we use the signed-rank test when the population is normal?

This section compares the three tests of the location parameter (t-test, sign test, signed-rank test). Their estimators and intervals for θ also differ: the t-test gives $\bar{X}_n$ with interval $\bar{X}_n \pm z_{\alpha/2}\hat{\sigma}/\sqrt{n}$; the signed-rank test gives median$\{W_{(ij)}\}$ with interval $(W_{(l)}, W_{(u)})$; the sign test gives median$\{X_i\}$ with interval $(X_{(l)}, X_{(u)})$. Which is best?

Under a parametric model (e.g. normal) there is an optimal estimator ($\bar{X}_n$ under normality) and an optimal test (the t-test). Under nonparametric models the family is so large that no method is optimal in every case: the t-test is best for the normal but may not be for the uniform. So we compare efficiencies in several representative populations.

2.3.1 Comparing inference methods

Comparing estimators. Which of $\bar{X}_n$, median$\{W_{(ij)}\}$, median$\{X_i\}$ is the best estimator of the population mean θ (or median θ)? The usual criterion is the mean squared error (MSE); the smaller the MSE, the better the estimator. The MSE of an estimator $\hat{\theta}$ of θ is

$$\mathrm{MSE}(\hat{\theta}, \theta) = E(\hat{\theta} - \theta)^2.$$

If $\hat{\theta}$ is unbiased, $\mathrm{MSE}(\hat{\theta}, \theta) = \mathrm{Var}(\hat{\theta})$, so we compare variances or standard deviations (standard errors). The expectation $E(\cdot)$ in the MSE is with respect to the sampling distribution of $\hat{\theta}$, which is determined by the population (the distribution of X_i) and the sample size n.

Relative efficiency of two estimators. If the target is always θ, write $\text{MSE}(\hat{\theta}, \theta) = \text{MSE}(\hat{\theta})$. Smaller MSE is better, so define the **efficiency**

$$e(\hat{\theta}) = \frac{c}{\text{MSE}(\hat{\theta})},$$

where the constant c is the minimal $\text{MSE}(\hat{\theta})$ over the family, so that $e(\hat{\theta}) \in [0, 1]$; the closer to 1, the more efficient. To compare $\hat{\theta}_1$ and $\hat{\theta}_2$ use the **relative efficiency**

$$RE(\hat{\theta}_1, \hat{\theta}_2) = \frac{e(\hat{\theta}_1)}{e(\hat{\theta}_2)} = \frac{\text{MSE}(\hat{\theta}_2)}{\text{MSE}(\hat{\theta}_1)}. \tag{2.18}$$

If $RE(\hat{\theta}_1, \hat{\theta}_2) > 1$, $\hat{\theta}_1$ is more efficient; if < 1, $\hat{\theta}_2$ is. Like the MSE, the relative efficiency depends on the population and the sample size n.

Example 2.15

Estimating the mean of a normal population: the relative efficiency of $\bar{X}_{10}$ (from 10 observations) and $\bar{X}_{40}$ (from 40). If the population variance is σ^2 (unknown), $\text{MSE}(\bar{X}_n) = \sigma^2/n$, so

$$RE(\bar{X}_{40}, \bar{X}_{10}) = \frac{\text{MSE}(\bar{X}_{10})}{\text{MSE}(\bar{X}_{40})} = \frac{\sigma^2/10}{\sigma^2/40} = 4 :$$

quadrupling the sample size quadruples the efficiency.

Example 2.16

Estimating the mean μ of $N(\mu, \sigma^2)$: relative efficiency of the median $m_n \equiv \text{median}\{X_1, \ldots, X_n\}$ and the mean $\bar{X}_n$. The exact MSE of m_n depends on the distribution of order statistics and has no neat form, but for large n, $m_n \dot{\sim} N(\mu, \sigma^2\pi/(2n))$ is known, so $\text{MSE}(m_n) \approx \text{Var}(m_n) \approx \sigma^2\pi/(2n)$ and

$$RE(\bar{X}_n, m_n) = \frac{\text{MSE}(m_n)}{\text{MSE}(\bar{X}_n)} \approx \frac{\sigma^2\pi/(2n)}{\sigma^2/n} = \pi/2 = 1.57 : \tag{2.19}$$

for a normal population the sample mean is about 1.57 times as efficient as the sample median.

When the exact MSE cannot be computed for fixed n, as in Example 2.16, or when the relative efficiency depends on n, we compare efficiencies as $n \to \infty$, the **asymptotic relative efficiency** (ARE):

$$ARE(\hat{\theta}_1, \hat{\theta}_2) = \lim_{n\to\infty} RE(\hat{\theta}_1, \hat{\theta}_2). \tag{2.20}$$

For a normal population, by (2.19), $ARE(\bar{X}_n, m_n) = 1.57$: the sample mean is more efficient than the sample median.

Relative efficiency of two confidence intervals. Consider two intervals $\theta \in (L_1, U_1)$ and $\theta \in (L_2, U_2)$ with the same level $1 - \alpha$. If their coverage probabilities (the probability the interval contains the parameter) are exact, $P(\theta \in (L_i, U_i)) = 1 - \alpha$ for $i = 1, 2$, the shorter $\ell_i = U_i - L_i$, the more efficient. So define the efficiency of (L_i, U_i) as $e((L_i, U_i)) = 1/\ell_i^2$.[9]

If the point estimator $\hat{\theta}$ used to build an interval is asymptotically $N(\theta, \sigma_\theta^2/n)$, the asymptotic interval is $C_{\hat{\theta}} \equiv \hat{\theta} \pm z_{\alpha/2}\sigma_{\hat{\theta}}/\sqrt{n}$. Also $\text{MSE}(\hat{\theta}) \approx \sigma_\theta^2/n$, so the length of $C_{\hat{\theta}}$ is approximately

[9] Defining the efficiency as c/ℓ_i^2 for some $c > 0$ instead of $1/\ell_i^2$ does not change relative efficiencies. Interval efficiency is not standard; for convenience we define "efficiency" first and then look at relative efficiency. Using the reciprocal of the squared length makes it comparable to the relative efficiency of estimators.

$2z_{\alpha/2}\sigma_{\hat{\theta}}/\sqrt{n} \approx 2z_{\alpha/2}\sqrt{\text{MSE}(\hat{\theta})}$ and the efficiency is $e(C_{\hat{\theta}}) \approx \frac{1}{4z^2_{\alpha/2}\,\text{MSE}(\hat{\theta})}$. For two estimators $\hat{\theta}_1, \hat{\theta}_2$ with asymptotic intervals $C_{\hat{\theta}_1}, C_{\hat{\theta}_2}$, by (2.18) and (2.20),

$$ARE(C_{\hat{\theta}_1}, C_{\hat{\theta}_2}) = \lim_{n\to\infty} \frac{e(C_{\hat{\theta}_1})}{e(C_{\hat{\theta}_2})} = \lim_{n\to\infty} \frac{\text{MSE}(\hat{\theta}_2)}{\text{MSE}(\hat{\theta}_1)} = ARE(\hat{\theta}_1, \hat{\theta}_2) :$$

the ARE of two asymptotic intervals equals the ARE of the corresponding estimators.

Comparing tests. From a random sample $X_1, \ldots, X_n$ test $H_0 : \theta = \theta_0$ against $H_1 : \theta < \theta_0$ for the median θ. If the population is normal, the median is the mean and the t-test controls the type I error at exactly the given level: $P(\sqrt{n}(\bar{X}_n - \theta_0)/s_n < -t_{\alpha,n-1}) = \alpha$. If the population is not normal, the type I error can exceed the level. Some examples of type I error and size of parametric tests in various populations:

Example 2.17

Type I error of the t-test of $H_0 : \theta = 0$ when the population with median $\theta = 0$ is Uniform$(-1, 1)$. By the CLT, for large n the statistic $T_n = \sqrt{n}(\bar{X}_n - \theta_0)/s_n$ is approximately standard normal, so $P(T_n < -t_{\alpha,n-1}) \approx \alpha$ and the type I error is approximately controlled. For very small n the type I error can be approximated by a simple simulation: with $n = 5$ at $\alpha = 0.05$, the type I error is $P(T_5 < -t_{0.05,4}) = 0.056$, slightly above the level. With $n = 10$ the distribution of T_n is almost standard normal and the type I error is close to 0.05.

Example 2.18

Type I error of the t-test of $H_0 : \theta = 0$ when the population with median $\theta = 0$ is the slash distribution, whose tails are much longer than normal. ($Z \sim N(0, 1)$ and $U \sim$ Uniform$(0, 1)$ independent: $X = Z/U$ is *slash*-distributed, and $E(X)$ does not exist.) For small n, e.g. $n = 10$ at $\alpha = 0.05$, the type I error of the t-test is about 0.029 (99% CI $(0.028, 0.030)$), a test more conservative than the level (rejecting H_0 less often than necessary). The CLT does not apply, so there is no guarantee that the type I error converges to 0.05 as n grows: for $n = 100$ it is still about 0.031, well below the level.

Example 2.19

Now an asymmetric population. The median and mean differ, so the t-test cannot be used for hypotheses about the median. We consider (a) using the t-test anyway and (b) setting new hypotheses about the population mean and running the t-test at $\alpha = 0.05$; as a simple asymmetric population take the exponential.

(a) The population is exponential with median $\theta = \log(2)$; test $H_0 : \theta = \log(2)$ against $H_1 : \theta < \log(2)$ with the t-test. The exponential with mean $\mu = 1$ has median $\theta = \mu \log(2) = \log(2) < 1$, so the statistic $t_n = \sqrt{n}(\bar{X}_n - \log(2))/s_n$ diverges to infinity instead of converging to the standard normal. The type I error $P(t_n < -t_{\alpha,n-1}) \approx 0$.

(b) The population is exponential with mean $\mu = 1$; test $H_0 : \mu = 1$ against $H_1 : \mu < 1$ with the t-test. By the CLT t_n is approximately standard normal, but because of the skewness of the exponential the approximation is good only for very large n (convergence to normality is slow). At $n = 100$, t_n is not close enough to normal, and the type I error is $P(t_{100} < -t_{\alpha,n-1}) = 0.074 > \alpha$, not controlled. If instead of $-t_{\alpha,n-1} \approx -z_\alpha \approx -1.66$ we use $c = -1.925$, wrong rejections become less frequent and the type I error $P(t_{100} < -1.92) \approx 0.049 < \alpha$ is controlled.

Unlike these examples, nonparametric tests control the type I error exactly (or conservatively) for every population. Two tests are compared by the principles:

1. both must control the type I error exactly or conservatively;
2. among tests satisfying the first principle, the one more likely to reject H_0 when H_1 is true (higher power) is better.

A test whose type I error exceeds the level cannot in principle be compared. As in Example 2.19(b), we may adjust the rejection region so that the type I error is controlled and then compare powers. Comparing powers requires the distribution of the test statistic when H_1 is true; but unlike H_0, H_1 comprises many cases.[10] E.g. the distribution of t_n differs between $\theta = \theta_0 - 1$ and $\theta = \theta_0 - 2$ even under normality. Comparing tests A and B, A may be more powerful in some alternatives and B in others; the power also depends on n and α. The best approach is to assume the most plausible alternatives in the given situation and compare the two powers there.

Example 2.20

Using a random sample of size n, test the median θ; earlier studies show the population is symmetric. Compare the t-test, the sign test and the signed-rank test in the cases:

(a) $n = 10$ (very small) and $n = 100$ (large enough);

(b) the population is normal $X \sim N(\theta, 1)$ (thin tails), Laplace with median θ, $X \sim$ Laplace$(\theta, 1)$ (heavy tails), or slash with median θ, $X = \theta + Z/U$ (very heavy tails).

Powers at $\alpha = 0.05$ for the six combinations (six alternative distributions), by simulation:

population	sample size	t-test	sign test	signed-rank test
Normal, $\theta = 1/4$	$n = 10$	17.8%	4.5%	15.3%
Normal, $\theta = 1/4$	$n = 100$	80.0%	61.2%	78.2%
Laplace, $\theta = 1/4$	$n = 10$	14.7%	5.3%	14.3%
Laplace, $\theta = 1/4$	$n = 100$	55.1%	70.3%	68.2%
Slash, $\theta = 1/2$	$n = 10$	8.7%	4.3%	12.7%
Slash, $\theta = 1/2$	$n = 100$	10.8%	60.3%	58.9%

For a normal population the t-test is the most powerful at every sample size (as is also known theoretically). With $n = 100$ the power of the signed-rank test is almost that of the t-test. This conclusion also holds for very thin-tailed populations (e.g. uniform centered at θ).
For heavy-tailed populations the conclusion depends on n. With small n the sign test's power drops sharply and the signed-rank and t-tests are more powerful. With large n the sign and signed-rank tests are much more powerful than the t-test. For the extremely heavy-tailed slash distribution (where the CLT fails) the t-test's power is very small, while for the Laplace the power differs little from nonparametric methods. Especially for very small samples ($n = 10$) the t-test's power is almost that of nonparametric methods, so it is acceptable there.

Asymptotic relative efficiency of two tests. As above, the comparison of two tests can change with n, and for a fixed alternative the power of a test increases with n. So another way of comparing two tests: at a given alternative $\theta = \Delta$, compare the sample sizes needed to reach the same power, e.g. $\beta = 80\%$. For a given population (e.g. slash), let N_t be the sample size for which the level-α

[10] $H_0 : \theta = \theta_0$ also comprises many cases. If the family is $\{N(\theta, 1) : \theta \in (-\infty, \infty)\}$ there is only one case under H_0, $N(\theta_0, 1)$; but for $\{N(\theta, \sigma^2) : \theta \in (-\infty, \infty), \sigma^2 > 0\}$ there are many, depending on σ^2. Under this parametric model the null distribution of t_n is the same whatever σ^2. For a nonparametric model $\{f$: distribution with median θ, $\theta \in (-\infty, \infty)\}$, the null distribution of a test statistic may depend on f even under H_0.

t-test "δ_t" has power β at $\theta = \Delta$, and N_{SR} the sample size for which the signed-rank test "δ_{SR}" has power β in the same situation. The relative efficiency of the t-test to the signed-rank test is defined as[11]

$$RE(\delta_t, \delta_{SR}) = \frac{N_{SR}}{N_t}.$$

In Example 2.20, for the slash distribution the t-test needed $N_t = 100$ for power $\beta = 10\%$, whereas the signed-rank test reaches $\beta = 10\%$ already with $N_{SR} = 10$, so the relative efficiency of the t-test to the signed-rank test is $N_{SR}/N_t = 10/100 = 0.1$ (the signed-rank test is 10 times as efficient). Remember that the relative efficiency depends on the population under the alternative.

The asymptotic relative efficiency is the relative efficiency as the alternative value Δ approaches θ_0. When Δ and θ_0 are very close, alternative and null are hard to distinguish and very large samples are needed. As $\Delta \to \theta_0$, $N_{SR} \to \infty$ and $N_t \to \infty$, and the limit of their ratio is the ARE. The ARE of the t-test to the signed-rank test is

$$ARE(\delta_t, \delta_{SR}) = \lim_{\Delta \to \theta_0} \frac{N_{SR}}{N_t}.$$

2.3.2 Asymptotic relative efficiency of the t-test, sign test and signed-rank test

We compare the AREs of the three tests in several representative families. If the population is not symmetric, only the sign test controls the type I error, so we consider symmetric populations only, where hypotheses about the median θ are hypotheses about the mean. We compare the uniform (almost no tails), the normal, the logistic (heavier tails but close to normal), the Laplace (heavy tails), and the slash (tails so heavy the mean does not exist) (Figure 2.4 of the book).

Each of the three tests corresponds to an estimator of θ: the t-test to $\bar{X}_n$, the signed-rank test to median$\{W_{(ij)}\}$, the sign test to median$\{X_i\}$. For large n the rejection regions are given by the asymptotic null distributions of the estimators, and when these are normal the ARE of two tests equals the ARE of the two estimators:

$$ARE(\delta_t, \delta_{SR}) = ARE(\bar{X}_n, \text{median}\{W_{(ij)}\}).$$

The table compares the AREs of the three tests (or estimators):

distribution	ARE(sign, t)	ARE(SR, t)	ARE(sign, SR)
Uniform	$1/3$	1	$1/3$
Normal	$2/\pi \approx 0.64$	$3/\pi \approx 0.954$	$2/3$
Logistic	$\pi^2/12 \approx 0.82$	$\pi^2/9 \approx 1.09$	$3/4$
Laplace	2	1.5	$4/3$
Slash	∞	∞[12]	1.09^*

The ARE of the CLT-based t-test relative to nonparametric methods worsens as the tails get heavier. For very heavy tails (Laplace and slash) the sign test has the highest ARE. The ARE of the signed-rank test relative to the t-test is at worst 0.954 (in fact at least 0.864 over all symmetric populations), so it is efficient for every population.

Where the table comes from (supplied)

Let the population have density f symmetric about θ, variance σ^2 (possibly infinite). As $n \to \infty$:

[11] This is called Pitman's relative efficiency.

[12] For the slash distribution, MSE$(\bar{X}_n) = \infty$, so the ARE relative to the t-test is ∞. (* For the slash, ARE(sign, SR) cannot be obtained as a ratio of the first two columns and is computed directly from the formulas below.)

- $\bar{X}_n \dot{\sim} N(\theta, \sigma^2/n)$ (CLT);
- the sample median $m_n \dot{\sim} N(\theta, \frac{1}{4nf(\theta)^2})$;
- the Hodges–Lehmann estimator median$\{W_{(ij)}\} \dot{\sim} N(\theta, \frac{1}{12n(\int f^2)^2})$.

Taking ratios of these variances,

$$ARE(\text{sign}, t) = 4\sigma^2 f(\theta)^2, \qquad ARE(SR, t) = 12\sigma^2\Big(\int f^2\Big)^2, \qquad ARE(\text{sign}, SR) = \frac{f(\theta)^2}{3(\int f^2)^2}.$$

Check: for $N(0,1)$, $f(0)^2 = 1/(2\pi)$ and $\int \phi^2 = 1/(2\sqrt{\pi})$, giving $2/\pi$, $3/\pi$, $2/3$. For Uniform$(-1,1)$, $\sigma^2 = 1/3$, $f(0) = 1/2$, $\int f^2 = 1/2$, giving $1/3$, 1, $1/3$. For the Laplace density $\frac{1}{2}e^{-|x|}$, $\sigma^2 = 2$, $f(0) = 1/2$, $\int f^2 = 1/4$, giving 2, $3/2$, $4/3$. For the logistic, $\sigma^2 = \pi^2/3$, $f(0) = 1/4$, $\int f^2 = 1/6$, giving $\pi^2/12$, $\pi^2/9$, $3/4$.
Sketch for the median. For continuous f with $f(\theta) > 0$, $P(m_n \le \theta + t/\sqrt{n}) = P(\#\{X_i \le \theta + t/\sqrt{n}\} \ge \lceil n/2 \rceil)$. The count is $B(n, p_n)$ with $p_n = F(\theta + t/\sqrt{n}) = \frac{1}{2} + f(\theta)t/\sqrt{n} + o(n^{-1/2})$, and the normal approximation to the binomial gives

$$P(m_n \le \theta + t/\sqrt{n}) \to \Phi(2f(\theta)t).$$

So $\sqrt{n}(m_n - \theta) \xrightarrow{d} N(0, 1/(4f(\theta)^2))$.